

University of Pisa
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Master's Degree in Mathematics

Algebraic Number Theory 3

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1. Reciprocity laws

The main goal of this course is to generalize the Legendre symbol and its reciprocity laws. change the motivation for the course, this one is quite bad.

Definition 1.0.1 (Legendre symbol)

Let p be a prime and $a \in \mathbb{Z}$ an integer coprime with p. The *Legendre symbol* is defined as

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } x^2 \equiv a \bmod p \text{ is solvable} \\ -1 & \text{otherwise} \end{cases}$$

Properties

- *(Euler)* The Legendre symbol is multiplicative in a and for $p > 2$ it holds
$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p.$$
- *(Gauss)* It satisfies the following reciprocity law: if $a \neq b$ are odd primes then
$$\left(\frac{a}{b}\right)\left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}.$$
- In particular we have
$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}, \qquad \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

By generalizing the symbol and its properties to any power studying the absolute Galois group turns out to be foundamental.

1.1. Hilbert's reinterpretation

Hilbert generalized the Legendre symbol, originally defined for integers and more generally for rational numbers, to the completions of $\mathbb{Q}$: the p-adic fields $\mathbb{Q}_p$ and the real field $\mathbb{Q}_\infty = \mathbb{R}$.

Definition 1.1.2 (Hilbert symbol)

For any $p \in \mathbb{P} \cup \{\infty\}$ let $a, b \in \mathbb{Q}_p$. The *Hilbert symbol* is defined as

$$(a,b)_p := \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ is solvable in } \mathbb{Q}_p \\ -1 & \text{otherwise} \end{cases}$$

Lemma 1.1.1

The following are equivalent:
1. The Hilbert symbol has value $(a,b)_p = 1$;
2. The equation $z^2 - ax^2 - by^2 = 0$ has a non-trivial solution;
3. The element $a \in \mathbb{Q}_p$ is the norm of an element in $\mathbb{Q}_p\left(\sqrt{b}\right)$;
4. The element $b \in \mathbb{Q}_p$ is the norm of an element in $\mathbb{Q}_p(\sqrt{a})$.

Here the norm $N : \mathbb{Q}_p(\sqrt{a}) \to \mathbb{Q}_p$ is the map

$$\begin{cases} \mathrm{id}_{\mathbb{Q}_p} & \text{if } a \in \mathbb{Q}_p^2 \\ x + y\sqrt{a} \mapsto x - y\sqrt{a} & \text{if } a \notin \mathbb{Q}_p^2 \end{cases}$$

In general if $L|K$ is a finite Galois extension with group G, the norm is defined as $N(x) = \sum_{\sigma \in G} \sigma x$ for each $x \in L$.

Proof: We only prove $(2) \Leftrightarrow (3)$ since the rest is trivial. When $b \in \mathbb{Q}_p^2$ both conditions are always true, from now on suppose $b \notin \mathbb{Q}_p^2$. If (x, y, z) is a solution to the equation, so is $(1, y/x, z/x)$ and we get $a = (z/x)^2 - b(y/x)^2 = N\left(z/x + \sqrt{b} \cdot y/x\right)$. Conversely, if $a = N\left(z + \sqrt{b} \cdot y\right)$ for some $y, z \in \mathbb{Q}_p$ then the touple $(1, y, z)$ is a solution to the equation. □

Proposition 1.1.1

If $p \neq 2, \infty$ then for any $a, b \in \mathbb{Z}$ we have

$$\begin{aligned}(a,b)_p &= (-1)^{\frac{p-1}{2} v_p(a) v_p(b)} \cdot \left(\frac{a}{p}\right)^{v_p(b)} \cdot \left(\frac{b}{p}\right)^{-v_p(a)} \\ &= \left[\overline{(-1)^{v_p(a)v_p(b)} \cdot a^{v_p(b)} \cdot b^{-v_p(a)}}\right]^{\frac{p-1}{2}}\end{aligned}$$

where $v_p(a) = \max\{n \in \mathbb{N} : p^n \mid a\}$ is the p-adic valuation of a and $a \mapsto \overline{a}$ is reduction modulo p.

In particular if $p \nmid a$ it simplifies to $(a, b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$.

Lemma 1.1.2

Let $a, b, c \in \mathbb{Q}_p^\times$. We have:
1. $(1, c)_p = 1$;
2. $(-c, c)_p = 1$;
3. $(a, c)_p = 1 \Longrightarrow (ab, c)_p = (a, c)_p (b, c)_p$;
4. $(c, c)_p = (-1, c)_p$.

Proof: Let $N : \mathbb{Q}_p(\sqrt{c}) \to \mathbb{Q}_p$ be the norm. We are going to repeatedly apply Lemma 1.1.1.
1. We have $1 = N(1)$.
2. If $c \in \mathbb{Q}_p^{\times 2}$ is a square, then $N = \mathrm{id}_{\mathbb{Q}_p}$, so $-c = N(-c)$. Otherwise $-c = N(c)$.
3. If a and b are norms we have

   $$a = N(x), b = N(y) \Longrightarrow ab = N(xy), a^{-1} = N(x^{-1}).$$

   Therefore if $(a, c)_p = 1$ (i.e. $a = N(x)$) then $b = N(y) \Leftrightarrow ab = N(xy)$.
4. Notice that by definition of Hilbert's symbol it is invariant under multiplication of either argument by a square. Hence

   $$(c, c)_p = (-c, c)_p (c, c)_p = \left(-c^2, c\right)_p = (-1, c)_p.$$

□

Lemma 1.1.3

If $p > 2$ and $u, v \in \mathbb{Z}_p^\times$ then $(u, v)_p = 1$.

Proof: Let $\overline{u} = u \bmod p$ and $\overline{v} = v \bmod p$, we claim $1 - \overline{u}x^2 - \overline{v}y^2 = 0$ has a trivial solution in $\mathbb{F}_p$. Indeed, since $\mathbb{F}_p^\times$ is cyclic of order $p-1$ it contains $\frac{p-1}{2}$ squares. Adding back 0 we get

$$\left|\left\{x^2 : x \in \mathbb{F}_p\right\}\right| = \frac{p+1}{2} \implies \left|\left\{\overline{u}x^2 : x \in \mathbb{F}_p\right\}\right| = \left|\left\{1 - \overline{v}y^2 : y \in \mathbb{F}_p\right\}\right| = \frac{p+1}{2}.$$

By a counting argument these two sets have an element in common, yielding the desired solution.

We conclude by applying Proposition 1.1.2 to lift the roots we found to $\mathbb{Z}_p$. □

Proposition 1.1.2 (Hensel's lemma)

If we take a polynomial $f \in \mathbb{Z}_p[x]$ and a p-adic number $a_0 \in \mathbb{Z}_p$ such that

$$f(a_0) \equiv 0 \bmod p, \qquad f'(a_0) \not\equiv 0 \bmod p$$

then there exist a p-adic number $a \in \mathbb{Z}_p$ such that $f(a) = 0$ and $a \equiv a_0 \bmod p$. In other words, under these conditions roots can be lifted from $\mathbb{F}_p$ to $\mathbb{Z}_p$.

Proposition 1.1.3

Let $p > 2$ and $a, b \in \mathbb{Q}_p^\times$. Write them as $a = p^\alpha u, b = p^\beta v$ with $u, v \in \mathbb{Z}_p$, we have

$$(a,b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)^{-\beta} \left(\frac{\overline{v}}{p}\right)^{\alpha}.$$

Proof: Since $(a,b)_p$ only depends on the classes $a, b \bmod \mathbb{Q}_p^{\times 2}$ we may assume $\alpha, \beta \in \{0,1\}$. By symmetry we only need to check 3 cases:

- $\boxed{\alpha = \beta = 0}$ Apply Lemma 1.1.3.
- $\boxed{\alpha = 1, \beta = 0}$ We need to show $(pu, v)_p = \left(\frac{\overline{v}}{p}\right)$. We have

  $$(pu,v)_p \overset{\text{Lemma 1.1.3}}{=} (pu,v)_p \cdot (u^{-1},v)_p = (p,v)_p.$$

  If $v \in \mathbb{Q}_p^{\times 2}$ is a square, then $(p,v)_p = (p,1)_p = 1 = \left(\frac{\overline{v}}{p}\right)$. Otherwise, the equation $z^2 - px^2 - y^2 = 0$ has no non-trivial solutions in $\mathbb{F}_p$. By homogeneity it has no non-trivial solutions in $\mathbb{Q}_p$ either, that is $(p,v)_p = -1$. By Proposition 1.1.2 we also have $\left(\frac{\overline{v}}{p}\right) = -1$ (if $\overline{v}$ was a square in $\mathbb{F}_p^\times$ it could be lifted to a square in $\mathbb{Z}_p^\times$).
- $\boxed{\alpha = \beta = 1}$ We need to show $(pu, pv)_p = (-1)^{\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right)$. We have

  $$(pu,pv)_p \overset{\text{Lemma 1.1.2}}{=} (pu,pv)_p(-pv,pv)_p = (-uv,pv)_p.$$

  Since $p \nmid uv$ we can apply the previous case and get

  $$(-uv,pv)_p = \left(\frac{-\overline{uv}}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right) = (-1)^{\frac{p-1}{2}}\left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right).$$

□

Corollary 1.1.1

When $p > 2$ we have $(a,c)_p(b,c)_p = (ab,c)_p$ for every $a, b, c \in \mathbb{Q}_p^\times$.

Proof: Let $a = p^\alpha u, b = p^\beta v, c = p^\gamma w$ with $u, v, w \in \mathbb{Z}_p^\times$. We have

$$\begin{aligned}(a,c)_p(b,c)_p &= (-1)^{\alpha\gamma\frac{p-1}{2}}\left(\frac{u}{p}\right)^{-\gamma}\left(\frac{w}{p}\right)^{\alpha}\cdot(-1)^{\beta\gamma\frac{p-1}{2}}\left(\frac{v}{p}\right)^{-\gamma}\left(\frac{w}{p}\right)^{\beta}\\ &= (-1)^{(\alpha+\beta)\gamma\frac{p-1}{2}}\left(\frac{uv}{p}\right)^{-\gamma}\left(\frac{w}{p}\right)^{\alpha+\beta}\\ &= (ab,c)_p\end{aligned}$$

□

Proposition 1.1.4

Let $a,b\in\mathbb{Q}_2^\times$ and write them as $a=2^\alpha u, b=2^\beta v$ with $u,v\in\mathbb{Z}_2^\times$. Define

$$\varepsilon(u) := \frac{u-1}{2} \bmod 2, \qquad \omega(u) := \frac{u^2-1}{8} \bmod 2.$$

The 2-adic Hilbert symbol has value

$$(a,b)_2 = (-1)^{\varepsilon(u)\varepsilon(v)+\alpha\omega(v)+\beta\omega(u)}.$$

It is a bimultiplicative function.

Theorem 1.1.1 (Hilbert's reciprocity law)

For any $a,b\in\mathbb{Q}^\times$ we have

$$\prod_p (a,b)_p = 1.$$

Proof: Hilbert's symbol is trivial for all but finitely many p, hence the product is well defined. Using multiplicativity in both entries we reduce to the case where both a and b are either -1 or prime.

- $\boxed{a=b=-1}$ *Exercise.*
- $\boxed{a=b=q}$ We reduce to $(q,q)_p=(-1,q)_p$. *Exercise.*
- $\boxed{a=2, b>2}$ If $p\notin\{2,b\}$ then $(2,b)_p=1$ by Proposition 1.1.3. Otherwise,

$$(2,b)_2 = (-1)^{\omega(b)} = \left(\frac{2}{b}\right) = (2,b)_b.$$

- $\boxed{a>b>2}$ If $p\notin\{2,a,b\}$ then $(a,b)_p=1$. Othewise,

$$(a,b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}, \qquad (a,b)_b = \left(\frac{a}{b}\right), \qquad (a,b)_a = \left(\frac{b}{a}\right),$$

  and by Gauss's reciprocity law we conclude.

□

Hilbert's reciprocity law incorporates Gauss's quadratic reciprocity. Indeed, if $a,b>0$:

- $p\nmid 2ab \implies (a,b)_p=1$;
- $2\nmid ab \implies (a,b)_2=(-1)^{\frac{a-1}{2}\frac{b-1}{2}}$;
- $(aa',b)=(a,b)(a',b)$.

Taking a,b as distinct primes recovers Gauss's law. Hilbert's and Gauss's reciprocity laws are in fact equivalent.

1.2. Hilbert's 9th problem

Next, Hilbert proceeded to generalize the result to n-powers in arbitrary number fields.

Let's consider at first the finite extensions $K|\mathbb{Q}$ contatining the n-th roots of unity $\mu_n \subset K$. Denote by $\mathcal{O}_K$ the ring of integers of K. If $\mathfrak{p} \subset \mathcal{O}_K$ is a prime ideal the goal is to define a generalized Hilbert symbol $(a,b)_\mathfrak{p} \in \mu_n$ such that Hilbert's reciprocity law holds, that is $\prod_\mathfrak{p} (a,b)_\mathfrak{p} = 1$ for every $a, b \in K^\times$.

Definition 1.2.3 (Generalized Hilbert symbol in the presence of roots of unity)

We set

$$(a,b)_\mathfrak{p} := \left[\overline{(-1)^{v_\mathfrak{p}(a)v_\mathfrak{p}(b)} \cdot a^{v_\mathfrak{p}(b)} \cdot b^{-v_\mathfrak{p}(a)}}\right]^{\frac{q-1}{n}}$$

where $\cdot \mapsto \bar{\cdot}$ is reduction $\bmod \mathfrak{p}$ and $q = |\mathcal{O}_K/\mathfrak{p}|$.

In particular if $\mathfrak{p} \nmid a$ it simplifies to $(a,b)_\mathfrak{p} = \overline{a^{\frac{q-1}{n}}}^{v_\mathfrak{p}(b)}$. If $v_\mathfrak{p}(b) = 1$, the value $(a,b)_\mathfrak{p}$ does not depend on the choice of b, and it corresponds to the unique root of unity congruent to $a^{\frac{q-1}{n}} \bmod \mathfrak{p}$.

Definition 1.2.4 (Generalized Legendre symbol)

If $\mathfrak{p} \nmid a$ pick any $b \in K^\times$ such that $v_\mathfrak{p}(b) = 1$, we set

$$\left(\frac{a}{\mathfrak{p}}\right) := (a,b)_\mathfrak{p}.$$

This generalizes the classical Legendre symbol, in fact

$$\left(\frac{a}{\mathfrak{p}}\right) \neq 1 \iff x^n = a \bmod \mathfrak{p} \text{ is solvable.}$$

Properties

- $(a,b)_\mathfrak{p} = 1 \iff b$ is a norm from $K_\mathfrak{p}(\sqrt[n]{a})$;
- $(a,b)_\mathfrak{p} = 1 \iff a$ is a norm from $K_\mathfrak{p}\left(\sqrt[n]{b}\right)$;
- $\overline{(a,b)_\mathfrak{p}}$ is multiplicative in both entries and anti-commutative.

When the roots of unity are not contained in the field $\mu_n \not\subset K_\mathfrak{p}$ the Hilbert symbol can still be defined but it won't depend on two elements $a, b \in K^\times$. Suppose for a moment $\mu_n \subset K_\mathfrak{p}, a \notin K_\mathfrak{p}^{\times n}$. The extension $K_\mathfrak{p}(\sqrt[n]{a})|K_\mathfrak{p}$ is Galois with cylic group, let $G_a := \mathrm{Gal}\big(K_\mathfrak{p}(\sqrt[n]{a})|K_\mathfrak{p}\big) \simeq \mathbb{Z}/n\mathbb{Z}$. There is an isomorphism

$$\mathrm{Hom}(G_a, \mu_n) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}, \qquad \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n$$

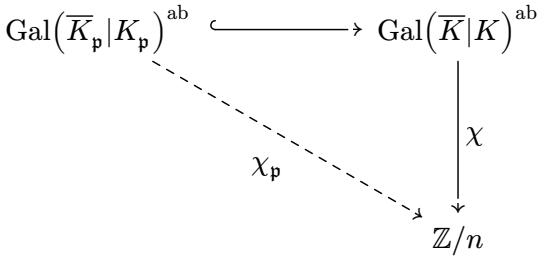
and therefore we can interpret a as the generator of this group of characters.

In general we define the Hilbert symbol

$$\mathrm{Hom}\Big(\mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big), \mathbb{Z}/n\mathbb{Z}\Big) \times K_\mathfrak{p}^\times \to \mathbb{Z}/n\mathbb{Z}, \qquad [\chi, b] \mapsto (\chi, b)_\mathfrak{p}.$$

In the case where $\mu_n \subset K_\mathfrak{p}$ the two symbols are effectively the same: let $\chi_a : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}}$, we have $(\chi_a, b)_\mathfrak{p} = (a, b)_\mathfrak{p}$.

Consider a character $\chi \in \mathrm{Hom}\big(\mathrm{Gal}\big(\overline{K}|K\big), \mathbb{Z}/n\mathbb{Z}\big)$, we shall see it induces a map $\chi_\mathfrak{p}$ via



Theorem 1.2.2 (Artin reciprocity law)

Fix an element $b \in K^\times$, we have

$$\sum_\mathfrak{p} (\chi_\mathfrak{p}, b) = 0.$$

The generalized Hilbert symbol defines a reciprocity law $K_\mathfrak{p}^\times \to \mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}}$ which is "almost" an isomorphism. In particular, for every n we have an isomorphism of finite groups $K_\mathfrak{p}^\times/n \xrightarrow{\sim} \mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}}/n$. To define such a map we shall contruct its dual

$$\begin{aligned} \mathrm{Hom}\Big(\mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}}/n, \mathbb{Z}/n\Big) &\longrightarrow \mathrm{Hom}\big(K_\mathfrak{p}^\times/n, \mathbb{Z}/n\big), \\ \chi &\longmapsto \big[b \mapsto (\chi, b)_\mathfrak{p}\big]. \end{aligned}$$

2. Cohomology of groups

3. p-adic number fields

Definition 3.0.5 (Brauer group)

Let k be a field and k_s be one of its separable closures. The *Brauer group* of k is

$$\mathrm{Br}(k) := H^2(k, k_s)$$

If $K|k$ is a separable subextension of $k_s|k$ we can define the *relative Brauer group* of the extension as

$$\mathrm{Br}(K|k) := \ker\left(\mathrm{Br}(k) \xrightarrow{\mathrm{Res}} \mathrm{Br}(K)\right)$$

The elements of $\mathrm{Br}(k)$ correspond to the finite dimensional division algebras D with center k. In this correspondence, the subgroup $\mathrm{Br}(K|k)$ gives those D for which $D \otimes_k K \simeq M_{n\times n}(K)$, where $n^2 = \dim_k D$.

Lemma 3.0.4

1. If $K|k$ is a Galois extension with group G then it holds $\mathrm{Br}(K|k) \simeq H^2(G, K^\times)$;
2. The Brauer group $\mathrm{Br}(k)$ is the union of all the subgroups $\mathrm{Br}(K|k)$ for finite Galois extensions $K|k$;
3. If G is finite and cyclic it holds $\mathrm{Br}(K|k) \simeq \frac{k^\times}{N_{K|k}(K^\times)}$.

Here $N_{K|k} : K^\times \to k^\times$ denotes the norm of the field extension. When $K|k$ is finite and Galois it has value

$$N_{K|k}(a) = \prod_{\sigma\in \mathrm{Gal}(K|k)} \sigma a$$

for every $a \in K^\times$.

Proof:

1. From Hilbert's theorem 90 we have $H^1(G, K^\times) = 0$. As seen in # this makes the following inflation-restriction sequence exact:

   $$0 \longrightarrow H^2(G, K^\times) \xrightarrow{\mathrm{Inf}} H^2(k, k_s) \xrightarrow{\mathrm{Res}} H^2(K, k_s),$$

   and by definition $\mathrm{Br}(K|k) = \ker \mathrm{Res} = H^2(G, K^\times)$.
2. From the previous point we deduce

   $$\mathrm{Br}(k) = \varinjlim H^2(G, K^\times) \overset{(1)}{=} \varinjlim \mathrm{Br}(K|k) = \bigcup \mathrm{Br}(K|k).$$
3. As seen in # when the Galois group G is cyclic we have

   $$\mathrm{Br}(K|k) \overset{(1)}{=} H^2(G, K^\times) = \frac{(K^\times)^G}{N_{K|k}(K^\times)} = \frac{k^\times}{N_{K|k}(K^\times)}.$$

□

Example 3.0.1

0. When the field k is complete we have $\mathrm{Br}(k) = 0$ since the Galois action on it is trivial.
1. Let's compute the Brauer group of the real numbers:

$$\mathrm{Br}(\mathbb{R}) = \mathrm{Br}(\mathbb{C}|\mathbb{R}) = H^2(\mathrm{Gal}(\mathbb{C}|\mathbb{R}), \mathbb{C}^\times) = \frac{\mathbb{R}^\times}{N_{\mathbb{C}|\mathbb{R}}(\mathbb{C}^\times)} = \frac{\mathbb{R}^\times}{\mathbb{R}^\times_{>0}} \simeq \mathbb{Z}/2\mathbb{Z}.$$

2. When $k = \mathbb{F}_q$ is a finite field the Brauer group vanishes: $\mathrm{Br}(\mathbb{F}_q) = 0$. By the lemma 2) it is enough to show that $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$ for every $r > 0$. Recall the following facts:
   - The group $\mathrm{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)$ is cyclic, generated by the frobenius $a \mapsto a^q$. In particular the norm has value
   $$N(a) = a^{\frac{q^r-1}{q-1}};$$
   - The multiplicative group $\mathbb{F}_{q^r}^\times$ is cyclic.

   Let ω be a generator of $\mathbb{F}_{q^r}^\times$, its norm $N(\omega)$ by the above calculation has order $q-1$ and therefore generates $\mathbb{F}_q$. This shows N is surjective and by lemma 3) we deduce $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$.

3.1. The path to $K|\mathbb{Q}_p$

The main goal of this section is proving that if $K|\mathbb{Q}_p$ is a p-adic number field then its Brauer group is $\mathrm{Br}(K) = \mathbb{Q}/\mathbb{Z}$.

Definition 3.1.6 (Discretely valuated fields)

A *discretely valuated field* is a field equipped with a discrete valuation $v : K \to \mathbb{Z} \cup \{\infty\}$, i.e. it holds
- $v(x) = \infty \iff x = 0$;
- $v(xy) = v(x)v(y)$;
- $v(x+y) \geq \min\{v(x), v(y)\}$.

Moreover we say K is *complete* if it complete with respect to the norm $\|x\|_v = e^{-v(x)}$. The *valuation ring* of K is defined as

$$\mathcal{O}_K := \{x \in K : v(x) \geq 0\},$$

it is a discrete valuation ring with maximal ideal $M_K := \{x \in K : v(x) > 0\}$. The *residue field* of K is the quotient $k(v) := \mathcal{O}_K/M_K$.

Fact

The ideal M_k is principal and it is generated by any element $\pi \in K$ with valuation $v(\pi) = 1$, such an element is called a *uniformiser* for $\mathcal{O}_K$. Since M_K is maximal, $k(v) = \mathcal{O}_K/(\pi)$ is indeed a field.

Fact

The field K is complete if and only if its valuation ring $\mathcal{O}_K$ is a complete DVR, that is

$$\mathcal{O}_K \simeq \varprojlim_i \frac{\mathcal{O}_K}{M_K^i}.$$

Example 3.1.2

1. The p-adic field $\mathbb{Q}_p$ is complete as $\mathbb{Q}_p = \mathrm{Frac}(\varprojlim \mathbb{Z}/p^i\mathbb{Z})$.
2. The Laurent series $k((t))$ form a complete field as $k((t)) = \mathrm{Frac}(\varprojlim k[\![t]\!]/(t^i))$.

Fact

If (K, v) is a complete discretely valuated field and $L|K$ is a finite separable extension, the valuation can be extended uniquely to L via

$$w = \frac{1}{[L:K]} v \circ N_{L|K} : L^\times \to \frac{1}{[L:K]}\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}.$$

The discretely valuated field (L, w) is complete itself and its valuation ring $\mathcal{O}_L$ is the integral closure of $\mathcal{O}_K$ inside L.

Definition 3.1.7 (ramification and residue class)

If $L|K$ is a finite separable extension of complete discretely valuated fields, the associated *ramification index* is $e(L|K) := [w(L^\times) : v(K^\times)]$ and the *residue class degree* is $f(L|K) := [k(w) : k(v)]$.

We say $L|K$ is *unramified* if $k(w)|k(v)$ is separable and $e = 1$; and $L|K$ is *totally ramified* if $f = 1$.

From now on we assume that K is a complete discretely valuated field.

Facts

- When $k(w)|k(v)$ is separable it holds $[L:K] = e(L|K) \cdot f(L|K)$.
- If $L|K$ is finite and unramified there exists a unique extension $L \supset M \supset K$ such that $M|K$ is unramified and $L|M$ is totally ramified.
- *Towers property:* If both $L|K, L'|L$ are finite and unramified, $L'|K$ is also finite and unramified.
- *Shift property:* If $L|K$ is finite and unramified, $K'|K$ is finite and separable; the extension $LK'|K'$ is finite and unramified.

Definition 3.1.8 (Unramified closure)

We set the maximal unramified extension of K as

$$K_{\mathrm{nr}} = \bigcup \{\overline{K} \supset L \supset K : L|K \text{ is finite and unramified}\}.$$

The maximal unramified extension K_{nr} is a dicretely valuated field, with valuation extending that of K. Its residue field is the separable closure $k(v)_s$, however in general it is not a complete field.

Suppose $L|K$ is a finite Galois extension (this can be generalized to infinite Galois extensions) and let $G = \mathrm{Gal}(L|K)$. The action of G on L is compatible with the valuation (this is a consequence of the uniqueness of valuation extensions), meaning $\mathcal{O}_L$ and M_L are G-invariant. Every element $\sigma \in G$ induces an automorphism

$$\overline{\sigma} : \frac{\mathcal{O}_L}{M_L} \to \frac{\mathcal{O}_K}{M_K}, \qquad \overline{x} \mapsto \overline{\sigma(x)},$$

giving a homomorphism on the Galois group of the residue fields extension

$$\rho : \mathrm{Gal}(L|K) \to \mathrm{Gal}(k(w)|k(v)).$$

By Hensel's lemma this is surjective, moreover its kernel is given by the *inertia subgroup* $I(L|K)$ and it is known that $|I(L|K)| = e(L|K)$. In particular when the extension is unramified (i.e. $e = 1$) the map ρ is an isomorphism.

Notice that by a similar argument if $L|K$ is a finite unramified extension then $L|K$ is Galois if and only if the residue field extension $k(w)|k(v)$ is Galois.

Example 3.1.3

Suppose the residue field $k(v)$ is finite. Since finite extension on finite fields are Galois, every finite unramified extension $L|K$ is a cyclic Galois extension. This means that for every $n \mid [L : K]$ there is a unique unramified subextension $L \supset K_n \supset K$ of degree $[K_n : K] = n$. In other words if we fix a closure $\overline{K}$ for every n there is a unique unramified Galois extension $\overline{K} \supset K_n \supset K$ of degree n.

Definition 3.1.9 (Units and principal units groups)

The *units group* of K is the subgroup $\mathcal{U}_K := \{a \in K^\times : v(a) = 0\}$. The *$i$-th group of principal units* is $\mathcal{U}_K^{(i)} := \{a \in K^\times : v(a-1) \geq i\}$.

There is an exact sequence of groups

$$1 \longrightarrow \mathcal{U}_K \longrightarrow K^\times \xrightarrow{v} \mathbb{Z} \longrightarrow 1,$$

the units group can be interpreted as the kernel of the valuation map. Moreover there is a surjection $\mathcal{U}_K \twoheadrightarrow k(v)^\times$ given by $u \mapsto u \bmod M_K$ and the principal units group can be viewed as its kernel:

$$\mathcal{U}_K^{(1)} = \ker(\mathcal{U}_K \longrightarrow k(v)^\times).$$

Notice that $\bigcap_i \mathcal{U}_K^i = \{1 + a \in K^\times : v(a) \geq i \text{ for each } i \geq 1\} = \{1\}$. The map $\mathcal{U}_K^{(i)} \to M_K^i$ given by $u \mapsto u \bmod M_K^{i+1}$ induces an isomorphism

$$\frac{\mathcal{U}_K^{(i)}}{\mathcal{U}_K^{(i+1)}} \xrightarrow{\sim} \frac{M_K^i}{M_K^{i+1}} \simeq \frac{\pi^i \mathcal{O}_K}{\pi^{i+1}\mathcal{O}_K} \simeq \frac{\mathcal{O}_K}{(\pi)} \simeq k(v).$$

We have therefore constructed a filtration (consisting of an infinite number of elements)

$$\mathcal{U}_K \underset{k(v)^\times}{\supset} \mathcal{U}_K^{(1)} \underset{k(v)^+}{\supset} \mathcal{U}_K^{(2)} \underset{k(v)^+}{\supset} \ldots \supset \{1\},$$

this filtration can be useful to prove properties of the multiplicative group $K^\times$.

Example 3.1.4

Consider a finite p-adic extension $K|\mathbb{Q}_p$. The residue field $k(v)$ is a finite extension of $\mathbb{F}_p$, hence it is of the form $k(v) = \mathbb{F}_q$ where $q = p^r, r = \big[K : \mathbb{Q}_p\big]$. The exact sequence

$$1 \longrightarrow \mathcal{U}_K^{(1)} \longrightarrow \mathcal{U}_K \longrightarrow k(v)^\times \to 1$$

splits by lifting $k(v)^\times$ to the $q-1$ roots of unity in $\mathcal{U}_K$ via $k(v)^\times \to \mu_{q-1}$. This gives the decomposition

$$K^\times \simeq \mathbb{Z} \times \mathcal{U}_K \simeq \mathbb{Z} \times \mu_{q-1} \times \mathcal{U}_K^{(1)}.$$

It can be shown that $\mathcal{U}_K^{(1)}$ itself can be decomposed as $\mathcal{U}_K^{(1)} \simeq \mu_{p^s} \times \mathbb{Z}_p^r$, where p^s is the number of p-power roots of unity in K.

We deduce $K^\times / K^{\times n}$ is a finite abelian group.

3.2. Computing $\mathrm{Br}(K)$ in the finite residue field case

The goal of this section is to show that whenever K is a complete discretely valuated field with finite residue field (as in a finite extension $K|\mathbb{Q}_p$), the Brauer group is given by $\mathrm{Br}(K) \simeq \mathbb{Q}/\mathbb{Z}$. We divide our goal in two steps: proving $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$ and proving $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathrm{Br}(K)$.

3.2.1. Showing $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$

Lemma 3.2.5

Let $L|K$ be a finite unramified Galois extension and let $G = \mathrm{Gal}(L|K)$. There is a split short exact sequence

$$0 \longrightarrow H^2(G, \mathcal{U}_L) \longrightarrow H^2(G, L^\times) \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

Proof: The short exact sequence

$$1 \longrightarrow \mathcal{U}_L \longrightarrow L^\times \xrightarrow{v} \mathbb{Z} \longrightarrow 0$$

splits via $s : \mathbb{Z} \to L^\times, 1 \mapsto \pi$ where π is a uniformiser of $\mathcal{O}_L$. Since the extension is unramified, we can choose $\pi \in K$, so that the section s is G-equivariant. Thus s induces a map on cohomology (that we still denote s) and the exact sequence

$$0 = H^1(G, \mathbb{Z}) \longrightarrow H^2(G, \mathcal{U}_L) \longrightarrow H^2(G, L^\times) \longrightarrow H^2(G, \mathbb{Z}) \longrightarrow 0$$

also splits via the section $s : H^2(G, \mathbb{Z}) \to H^2(G, L^\times)$ (indeed, the existence of the section guarantees surjectivity of the right-hand map). It remains to justify that $H^1(G, \mathbb{Z}) = 0$: since the G-action on $\mathbb{Z}$ is trivial we have $H^1(G, \mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Z})$ and every homomorphism from a finite group to $\mathbb{Z}$ is trivial.

It remains to identify $H^2(G, \mathbb{Z})$ with $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$. Consider the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0,$$

the associated long sequence in cohomology gives

$$0 = H^1(G, \mathbb{Q}) \longrightarrow H^1(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} H^2(G, \mathbb{Z}) \longrightarrow H^2(G, \mathbb{Q}) = 0.$$

As we've seen since $\mathbb{Q}$ is divisible its finite group cohomology vanishes in positive dimension. Since the G-action on $\mathbb{Q}/\mathbb{Z}$ is trivial we have $H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ and we conclude. □

Lemma 3.2.6

Let G be a finite group and let (A_j, φ_j) be an inverse system of G-modules indexed by $\mathbb{Z}_{>0}$ (in particular the transition maps are the $\varphi_j : A_{j+1} \to A_j$). For any i, if $H^i(G, A_1) = 0$ and $H^i(G, \ker \varphi_j) = 0$ for every j, then $H^i(G, \lim_{\leftarrow} A_j) = 0$.

Lemma 3.2.7 (Additive Hilbert's 90)

If K is a field and $L|K$ is a finite Galois extension with group G, then $H^i(G, L^+) = 0$ for every $i > 0$.

In particular from Lemma 3.2.6 we deduce $H^i(k, k_s) = 0$ for every $i > 0$.

Proof: Recall the *normal basis theorem*: there is an element $a \in L$ such that the set $\{\sigma a : a \in G\}$ forms a k-basis for L. We deduce there is an isomorphism of G-modules

$$k^+ \otimes_{\mathbb{Z}} \mathbb{Z}[G] \to L^+, \qquad \lambda_i \otimes \sigma_i \mapsto \lambda_i \sigma_{i(a)}.$$

Moreover there is a non-canonical isomorphism $k^+ \otimes_{\mathbb{Z}} \mathbb{Z}[G] \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], k^+)$ of k-vector spaces since they have the same (finite) dimension. The latter is the coinduced-G-module $M^G(k^+)$ and therefore by Shapiro's lemma $H^i(G, M^G(k^+)) = 0$. □

Proposition 3.2.5

Let $L|K$ be a finite unramified Galois extension with Galois group G. Let κ and λ denote the residue fields of K and L, respectively. The reduction map $\mathcal{U}_L \to \lambda^\times$ induces an isomorphism $H^2(G, \mathcal{U}_L) \xrightarrow{\sim} H^2(G, \lambda^\times)$.

Proof: The quotients $\mathcal{U}_L^{(1)}/\mathcal{U}_L^{(j)}$ form an inverse system of groups and since the principal units groups $\mathcal{U}_L^{(j)}$ have trivial intersection we get an isomorphism $\mathcal{U}_L^{(1)} \simeq \lim_{\leftarrow} \mathcal{U}_L^{(1)}/\mathcal{U}_L^{(j)}$. For every $j \geq 1$ there is an exact sequence

$$1 \longrightarrow \mathcal{U}_L^{j+1} \longrightarrow \mathcal{U}_L^{j} \longrightarrow \lambda^+ \longrightarrow 1 \Longrightarrow \lambda^+ \simeq \mathcal{U}_L^{(j)}/\mathcal{U}_L^{(j+1)}$$

For every $i > 0$, by Lemma 3.2.7 we have $H^i(G, \lambda^+) = 0$ and therefore $H^i\left(G, \mathcal{U}_L^{(j)}/\mathcal{U}_L^{(j+1)}\right) = 0$. Applying Lemma 3.2.6 yields $H^i\left(G, \mathcal{U}_L^{(1)}\right) = 0$. Now extract the cohomology long exact sequence from $1 \to \mathcal{U}_L^{(1)} \to \mathcal{U}_L \to \lambda^\times \to 1$, we get

$$0 = H^2\left(G, \mathcal{U}_L^{(1)}\right) \longrightarrow H^2(G, \mathcal{U}_L) \xrightarrow{\sim} H^2(G, \lambda^\times) \longrightarrow H^3\left(G, \mathcal{U}_L^{(1)}\right) = 0,$$

that is the isomorphism we desired. □

Corollary 3.2.2

We have a split short exact sequence

$$0 \longrightarrow \mathrm{Br}(\lambda|\kappa) \longrightarrow \mathrm{Br}(L|K) \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

In particular when κ is finite if we let $n = [L : K]$ then $\mathrm{Br}(L|K) \simeq \mathbb{Z}/n\mathbb{Z}$.

Proof: By Lemma 3.2.5 and Proposition 3.2.5 we get the desired sequence. When κ is finite, since the extension $\lambda|\kappa$ is itself finite (recall that $\mathrm{Gal}(\lambda|\kappa) \simeq \mathrm{Gal}(L|K)$ in our case) we have $\mathrm{Br}(\lambda|\kappa) = 0$. Moreover $n = [L : K] = |G|$ and therefore $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$. □

We can now apply the direct limit over all finite unramified Galois extension $L|K$ to the short exact sequence and get the following result:

Summary

If K is a complete dicretely valuated field with perfect residue field κ, then we have a split short exact sequence

$$0 \longrightarrow \mathrm{Br}(\kappa) \longrightarrow \mathrm{Br}(K_{\mathrm{nr}}|K) \longrightarrow \mathrm{Hom}_{\mathrm{cont}}\left(\mathrm{Gal}\left(\overline{K}|K\right), \mathbb{Q}/\mathbb{Z}\right) \longrightarrow 0.$$

If moreover κ is finite we get $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$.

3.2.2. Showing $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathrm{Br}(K)$

This fact is in general true for any complete discretely valuated field, however we are only going to prove it for fields K with a finite residue field.

Lemma 3.2.8

If K is a complete discretely valuated field and $L|K$ is a finite Galois extension with Galois group G, then the units group $\mathcal{U}_L$ contains a G-submodule $V_L^1 \subset \mathcal{U}_L$ of finite index such that $H^i(G, V_L^1) = 0$ for every $i > 0$.

Proof: From the *normal basis theorem* there is an $a \in L$ such that the set $\{\sigma a : \sigma \in G\}$ is a k-basis of L. Let π be a uniformiser of $\mathcal{O}_K$ (i.e. $v_K(\pi) = 1$), since G is finite there is a $r \gg 0$ such that $\pi^r \sigma a \in \mathcal{O}_L$ for every $\sigma \in G$. Define the $\mathcal{O}_K$-module $M = \langle \pi^r \sigma a : \sigma \in G \rangle_{\mathcal{O}_K}$, it is a submodule of $\mathcal{O}_L$. We know $\mathcal{O}_L$ is free of rank $[L : K]$ and by construction this also holds for M. By the structure theorem of finitely generated modules on PIDs the quotient $\mathcal{O}_L/M$ is finite (since it is torsion and finitely generated). This shows that $\pi^m \mathcal{O}_L \subset M$ for a suitable m, fix such an m. Define $V_L^j := 1 + \pi^{m+j} M \subset \mathcal{U}_L$ and consider the map

$$V_L^j \to \frac{M}{\pi M}, \qquad 1 + \pi^{m+j} a \mapsto a \bmod \pi M.$$

Its kernel is exactly V_L^{j+1}, therefore the map induces an isomorphism

$$\frac{V_L^j}{V_L^{j+1}} \xrightarrow{\sim} \frac{M}{\pi M} \simeq \frac{\mathcal{O}_K}{(\pi)} \otimes \mathbb{Z}[G] \simeq \kappa \otimes \mathbb{Z}[G].$$

By Shapiro's lemma, the cohomology group $H^i(G, \kappa \otimes \mathbb{Z}[G]) = 0$ since $\kappa \otimes \mathbb{Z}[G]$ is isomorphic to a coinduced-module. Moreover $V_L^1 \simeq \lim_{\leftarrow} V_L^1/V_L^j$ and by Lemma 3.2.6 we get $H^1(G, V_L^1) = 0$.

Finally, $V_L^1 \subset \mathcal{U}_L$ has finite index since it contains the subgroup $1+\pi^{2m+1}\mathcal{O}_L$, which has finite index in $\mathcal{U}_L$. □

Corollary 3.2.3

The cohomology group $H^i(G,\mathcal{U}_L)$ is finite for every $i>0$.

Proof: By the short exact sequence $1 \to V_L^1 \to \mathcal{U}_L \to \mathcal{U}_L/V_L^1 \to 1$ we extract the cohomology long exact sequence

$$0 = H^i(G,V_L^1) \longrightarrow H^i(G,\mathcal{U}_L) \xrightarrow{\sim} H^i(G,\mathcal{U}_L/V_L^1) \longrightarrow H^{i+1}(G,V_L^1) = 0,$$

Indeed, by the previous lemma $H^i(G,V_L^1)=0$ and the quotient $\mathcal{U}_L/V_L^1$ is finite. □

Lemma 3.2.9

Let G be a finite cyclic group and let A be a finite G-module. We have

$$\big|H^1(G,A)\big| = \big|H^2(G,A)\big|.$$

Proof: We have computed the cohomology of a cyclic group:

$$H^1(G,A) = \frac{{}_NA}{(\sigma-1)A}, \qquad H^2(G,A) = \frac{A^G}{NA},$$

where σ is a generator of G and $N=\sum_{\sigma\in G}\sigma$. We have two short exact sequences

$$0 \longrightarrow A^G \longrightarrow A \longrightarrow (\sigma-1)A \longrightarrow 0$$
$$0 \longrightarrow {}_NA \longrightarrow A \longrightarrow NA \longrightarrow 0$$

from which we deduce $\big|A^G\big||(\sigma-1)A| = |A| = |{}_NA||NA|$. □

Proposition 3.2.6

If K is a complete discretely valuated field and $L|K$ is a finite Galois extension with cyclic Galois group $G \simeq \mathbb{Z}/n\mathbb{Z}$, then the relative Brauer group has order $|\mathrm{Br}(L|K)| = n$.

Proof: We need to show that $H^2(G,L^\times)$ is finite of order n. From the short exact sequence $1 \to \mathcal{U}_L \to L^\times \to \mathbb{Z} \to 1$ we extract the cohomology long exact sequence

$$0 = H^1(G,\mathbb{Z}) \longrightarrow H^2(G,\mathcal{U}_L) \longrightarrow H^2(G,L^\times) \longrightarrow H^2(G,\mathbb{Z}) \longrightarrow H^3(G,\mathcal{U}_L) \longrightarrow H^3(G,L^\times) = 0.$$

Indeed since G is finite $H^1(G,\mathbb{Z})=0$ and since G is also cyclic $H^3(G,L^\times) = H^1(G,L^\times) = 0$ by Hilbert's 90. Since all the elements in the sequence have finite order we deduce

$$\big|H^2(G,L^\times)\big|\big|H^3(G,\mathcal{U}_L)\big| = \big|H^2(G,\mathcal{U}_L)\big|\big|H^2(G,\mathbb{Z})\big| \overset{\text{Lemma 3.2.9}}{\Longrightarrow} \big|H^2(G,L^\times)\big| = \big|H^2(G,\mathbb{Z})\big|.$$

Since $H^2(G,\mathbb{Z}) \simeq \mathrm{Hom}(G,\mathbb{Q}/\mathbb{Z}) \simeq \mathbb{Z}/n\mathbb{Z}$ we conclude. □